

My Project

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Chapter 1

Class Index

1.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

FShop	5
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Chapter 2

File Index

2.1 File List

Here is a list of all documented files with brief descriptions:

ACO.h	7
FShop.h	7
MakespanFunctions.h	7
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Chapter 3

Class Documentation

3.1 FShop Class Reference

Public Member Functions

- [FShop](#) (std::size_t jobs, std::size_t machines, unsigned int(*makespanfn)(std::vector< std::vector< unsigned int > > &, std::vector< std::size_t > &))
- void [updateMakespan](#) ()

Public Attributes

- std::vector< std::vector< unsigned int > > **f_shop**
- std::vector< std::size_t > **job_index_order**
- unsigned int **makespan**
- unsigned int(* **calcMakespan**)(std::vector< std::vector< unsigned int > > &fs, std::vector< std::size_t > &ji)

3.1.1 Constructor & Destructor Documentation

3.1.1.1 FShop()

```
FShop::FShop (
    std::size_t jobs,
    std::size_t machines,
    unsigned int(* makespanfn ) (std::vector< std::vector< unsigned int > > &, std::vector< std::size_t > &))
```

Constructor of [FShop](#)

Parameters

in	<i>jobs</i>	the number of jobs in a flow shop
in	<i>machines</i>	the number of machines in a flow shop

in	<i>makespan</i>	a function to calculate the makespan of a flow shop
out	<i>a</i>	flow shop initialized with every value as [job][machine] as 0

3.1.2 Member Function Documentation

3.1.2.1 updateMakespan()

```
void FShop::updateMakespan ()
```

Updates the makespan of a [FShop](#) using its makespan function

The documentation for this class was generated from the following files:

- FShop.h
- FShop.cpp

Chapter 4

File Documentation

4.1 ACO.h

```
00001 // Name: Harrison Ingram-Bate
00002 #ifndef ACO_H
00003 #define ACO_H
00004
00005 #include "FShop.h"
00006
00019 FShop ACO(FShop& fs,
00020           std::size_t num_ants,
00021           int iterations,
00022           float alpha,
00023           float beta,
00024           float evaporation,
00025           float q);
00026
00027 #endif
```

4.2 FShop.h

```
00001 // Name: Harrison Ingram-Bate
00002 #ifndef F_SHOP
00003 #define F_SHOP
00004
00005 #include <vector>
00006
00007 class FShop {
00008 public:
00017     FShop(std::size_t jobs,
00018           std::size_t machines,
00019           unsigned int (*makespanfn)(std::vector<std::vector<unsigned int>&, std::vector<std::size_t>&));
00023     void updateMakespan();
00024
00025     std::vector<std::vector<unsigned int>> f_shop;
00026     std::vector<std::size_t> job_index_order;
00027     unsigned int makespan;
00028     unsigned int (*calcMakespan)(std::vector<std::vector<unsigned int>& fs, std::vector<std::size_t>&
                                ji);
00029 };
00030
00031 #endif
```

4.3 MakespanFunctions.h

```
00001 // Name: Harrison Ingram-Bate
00002 #ifndef MAKESPAN_FUNCTIONS
00003 #define MAKESPAN_FUNCTIONS
00004
00005 #include <vector>
00006
```

```
00007 namespace makespan_functions{
00008
00009     unsigned int nonBlocking(std::vector<std::vector<unsigned int>& fs, std::vector<std::size_t>& ji);
00010
00011     unsigned int blocking(std::vector<std::vector<unsigned int>& fs, std::vector<std::size_t>& ji);
00012 }
00013
00014 #endif
```

4.4 NEH.h

```
00001 // Name: Harrison Ingram-Bate
00002 #ifndef NEH_H
00003 #define NEH_H
00004
00005 #include "FShop.h"
00006
00013 FShop NEH(FShop& fs);
00014
00015 #endif
```

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